Riya Singh

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Summary

Computer Science undergraduate specializing in Machine Learning, NLP, and Generative AI systems with hands-on experience in designing scalable AI architectures. Skilled in building hybrid Retrieval-Augmented Generation (RAG) pipelines, semantic search engines, and deploying ML-powered applications using FastAPI, React, and AWS cloud infrastructure. Strong foundation in Data Structures, Algorithms, probability, and applied AI system design.

Work Experience

AI/ML Intern - Labmentix Jun 2025 – Sep 2025

- Engineered and modeled 50k+ structured records, improving predictive performance by 12% through feature optimization and hyperparameter tuning.
- Designed complete ML workflow including preprocessing, scaling, validation, and model benchmarking across multiple algorithms.
- Implemented stratified validation, performance tracking (precision, recall, F1-score), and confusion matrix evaluation.
- Structured reproducible experimentation pipelines ensuring seamless model deployment.

Associate - Next Tech Lab (McCarthy Lab), SRM AP Apr 2025 – Jan 2026

- Contributed to ML, NLP, and Generative AI research discussions focusing on scalable AI system architectures.
- Assisted in experimentation, optimization strategies, and evaluation of model generalization techniques.
- Collaborated in academic technical reviews and solution prototyping sessions.

Projects

Medical Report Analyzer - Hybrid RAG System GitHub

Python, FAISS, SentenceTransformers, FastAPI, AWS

- Architected hybrid Retrieval-Augmented pipeline combining FAISS semantic indexing with rule-based medical entity extraction.
- Implemented PDF parsing, chunking strategies, embedding generation, similarity ranking, and structured parameter comparison.
- Automated classification of 15+ lab parameters (High/Low/Normal) and generated contextual medical summaries.
- Designed REST API backend for scalable integration and cloud deployment.

SafeNest - Discreet Domestic Safety Assistant GitHub

HTML, CSS, JavaScript, Geolocation API, MediaRecorder API, Google Apps Script

- Developed a responsive emergency safety web application enabling users to trigger SOS alerts, record audio/video evidence, and share live location.
- Implemented browser-based Geolocation API with reverse geocoding to detect city, area, and generate Google Maps navigation links.
- Integrated MediaRecorder API for automatic video evidence capture and cloud storage using Google Apps Script and Google Drive.
- Designed countdown-based emergency alert system triggering SMS, WhatsApp, and police contact options.

AI Resume Matcher - Full Stack NLP Application GitHub

React, FastAPI, NLP, SentenceTransformers, AWS

- Built embedding-based semantic similarity engine for resume-to-job matching.
- Developed FastAPI backend for PDF extraction, preprocessing, vector generation, and similarity scoring.
- Implemented skill-gap analysis module and ranking logic for recruiter insights.
- Structured application for cloud-ready deployment with modular API architecture.

Education

2023 – Present B.Tech in Computer Science & Engineering (AI & ML Specialization) at **SRM University, AP**

CGPA: 9.23 / 10.0

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning, Probability & Statistics, Computer Networks

Academic Distinction: Top 10% of cohort; consistent high academic performance.

Technical Skills

- **Languages:** Python, Java, JavaScript, SQL
- **Machine Learning:** Scikit-learn, Feature Engineering, Model Evaluation, Classification, Cross-Validation
- **Deep Learning:** TensorFlow, Keras, CNNs, Neural Networks, Model Optimization
- **NLP & GenAI:** RAG Systems, Embeddings, Semantic Search, Prompt Engineering
- **Web Development:** React, FastAPI, Flask, REST APIs, HTML, CSS
- **Cloud & Tools:** AWS (EC2, S3, IAM), Git, GitHub, FAISS